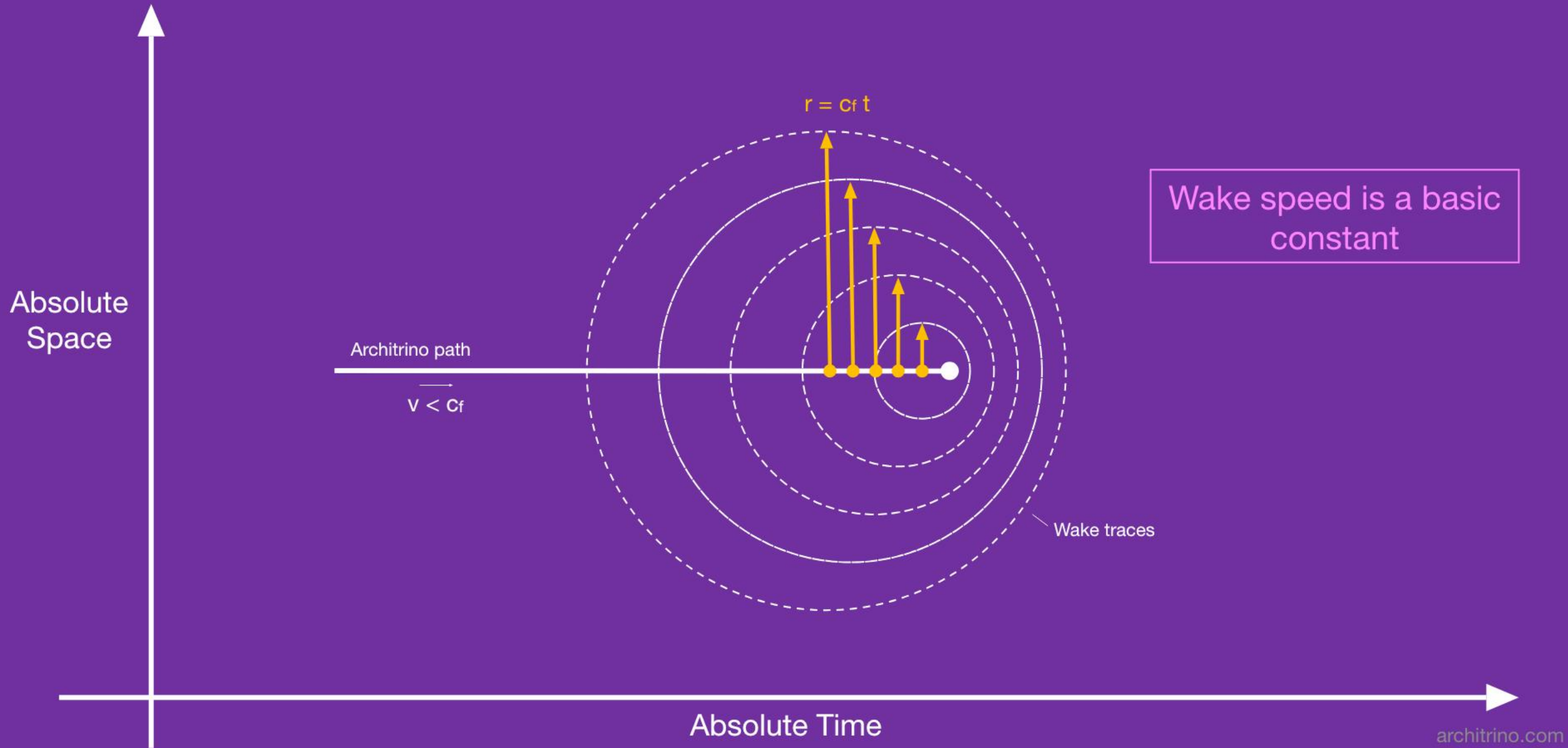


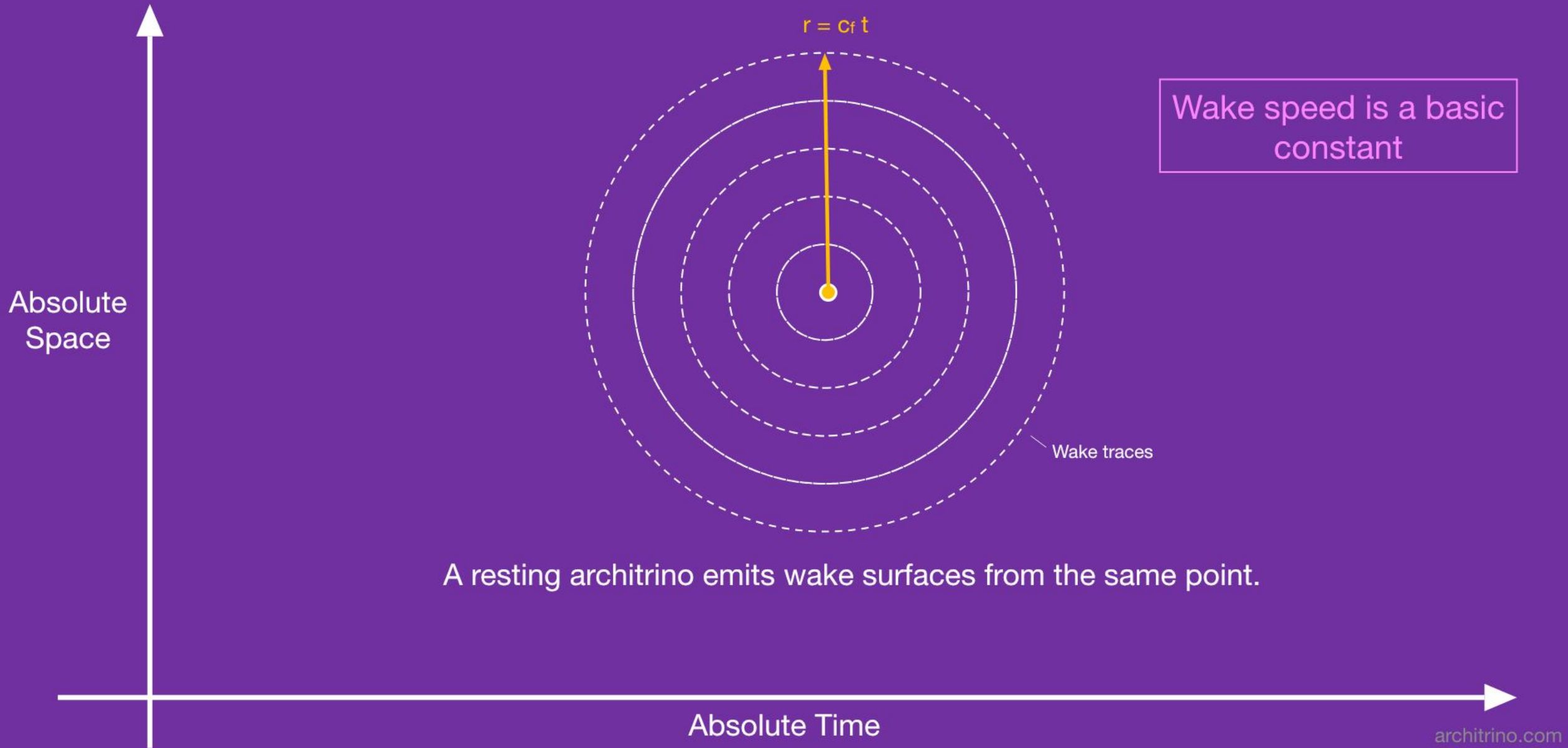
Architrino Paths and Causal Wakes

Causal wakes expand from each emission point

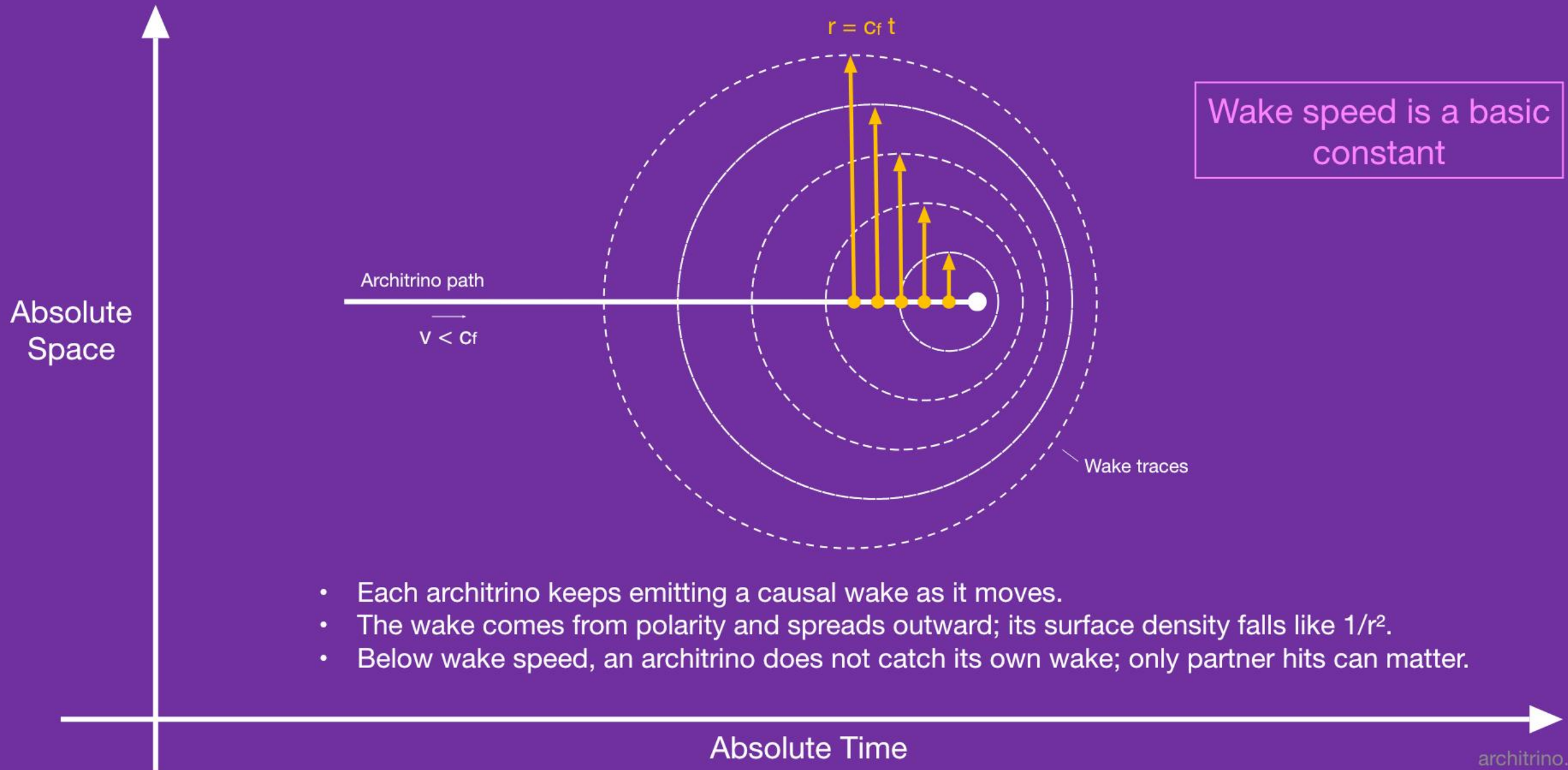
Each wake surface grows from the point where it was emitted



A resting architrino emits concentric wakes

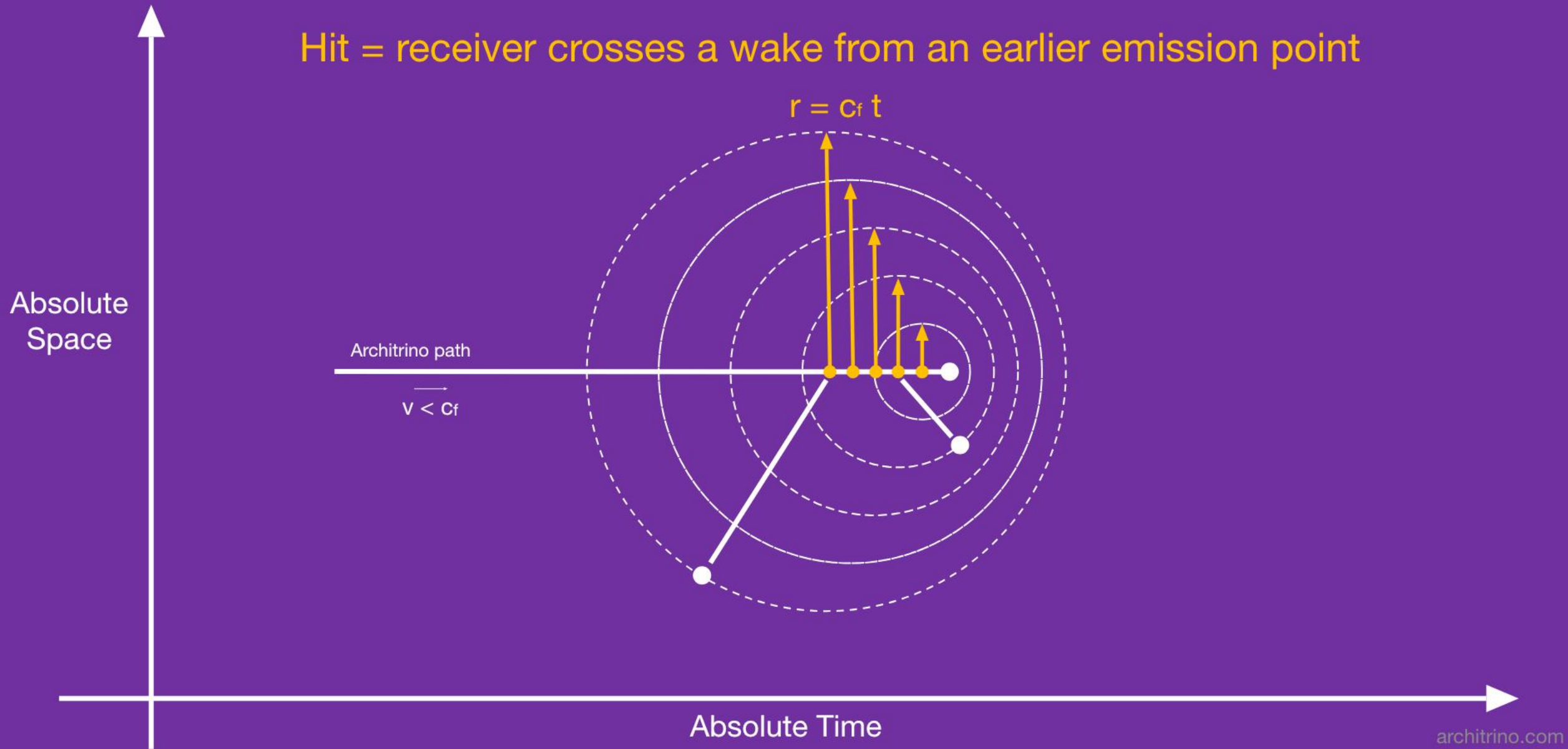


Below wake speed, the source has no self-hit

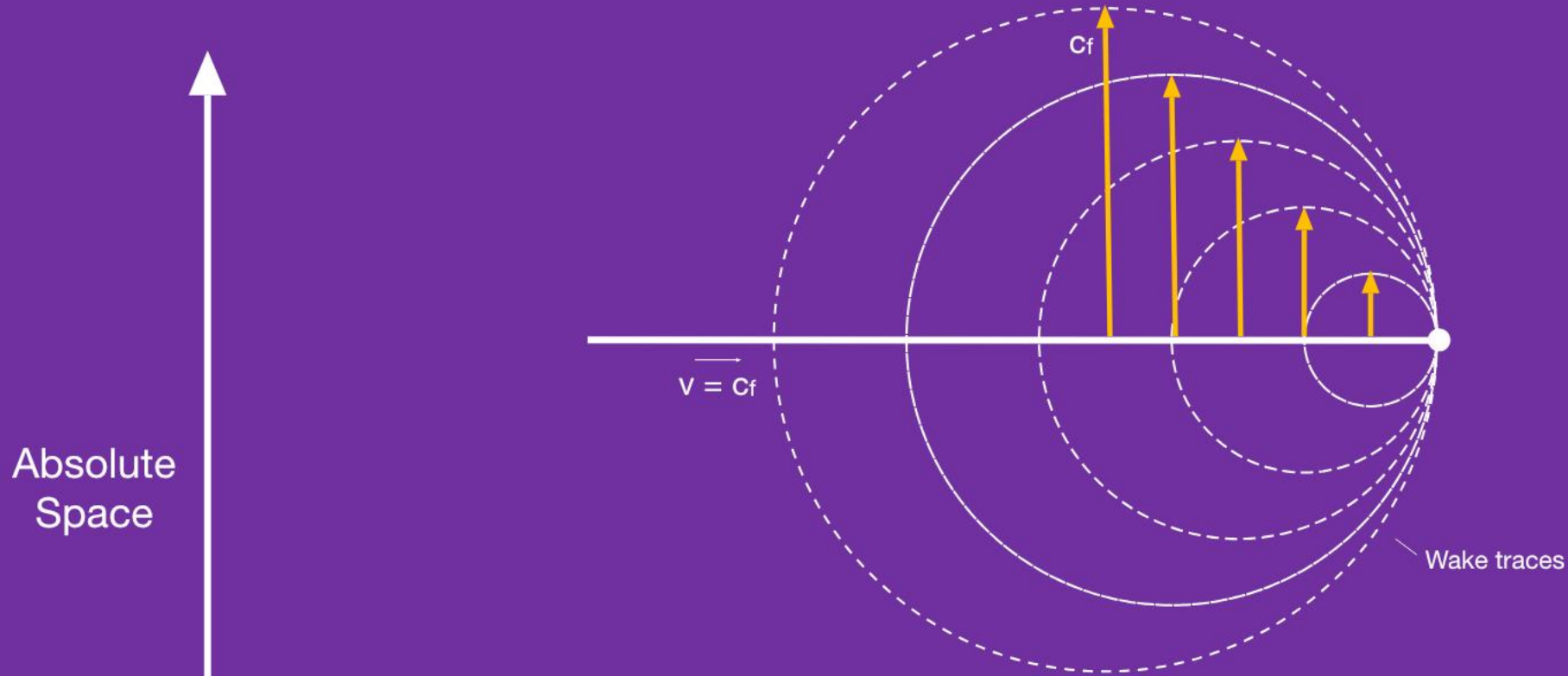


A hit happens where a receiver crosses a wake

Hit = receiver crosses a wake from an earlier emission point

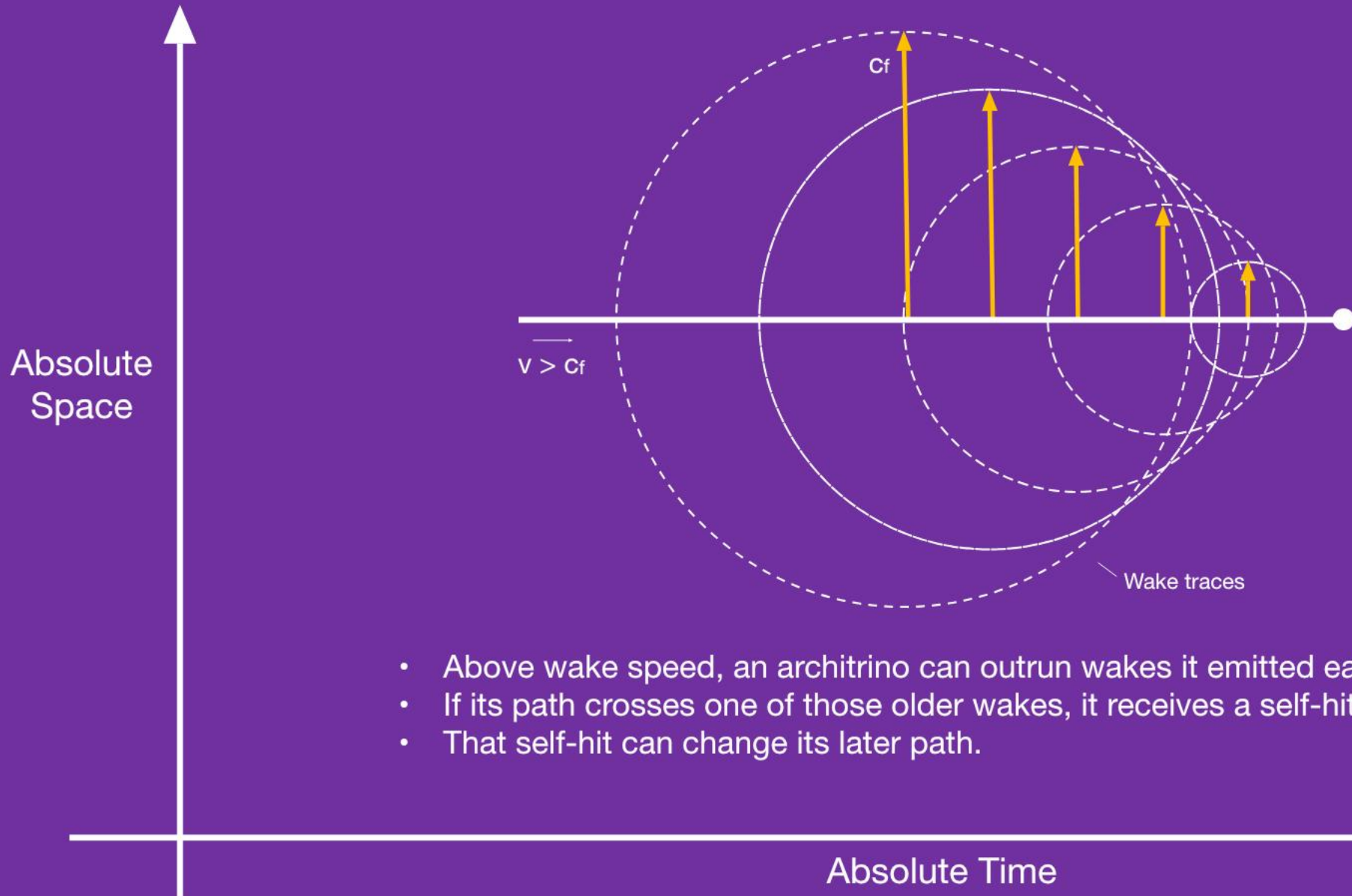


Wake speed marks the self-hit threshold



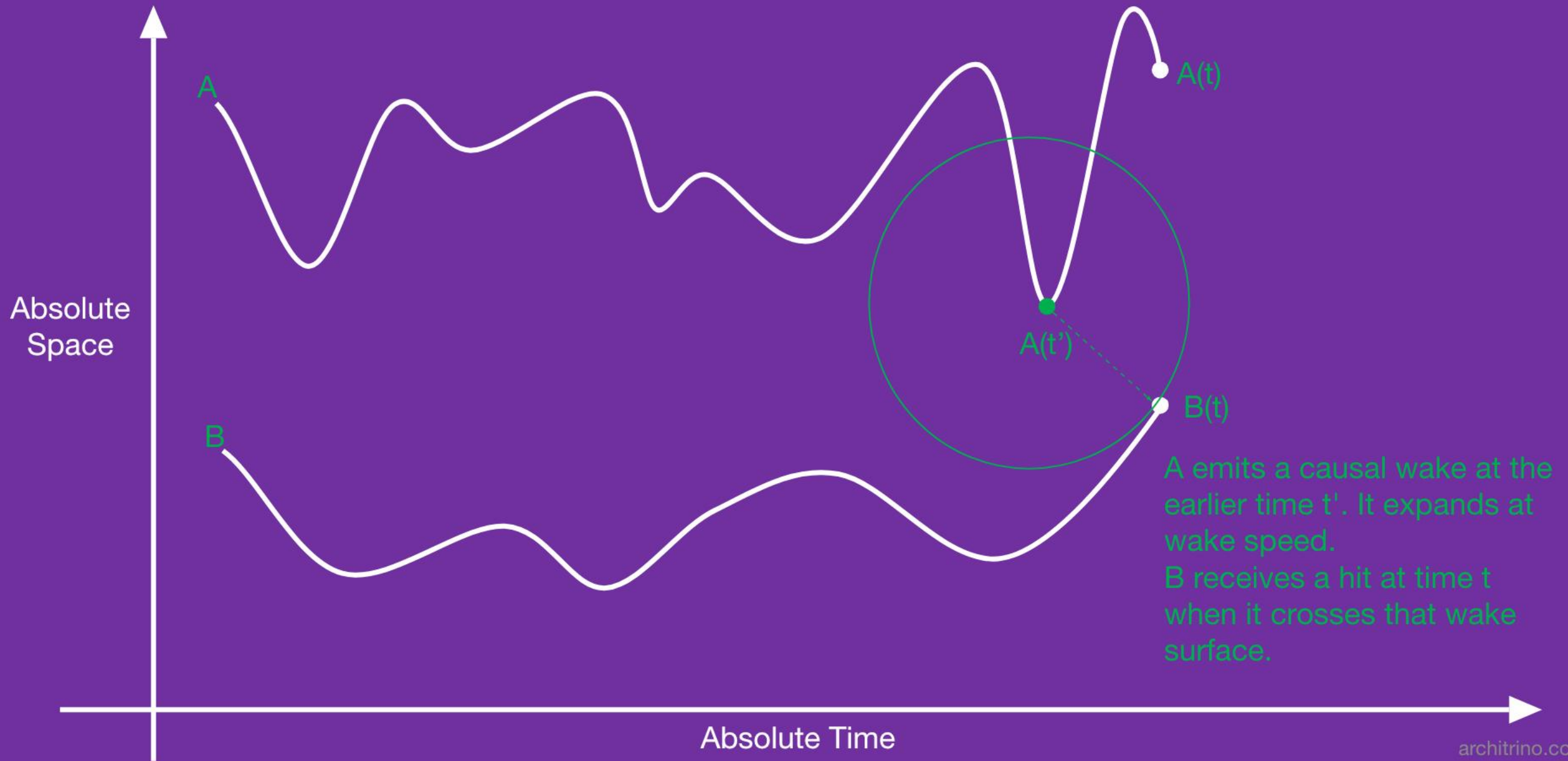
- At wake speed, the architrino reaches the self-hit threshold.
- At the threshold, it still has no retained self-hit.
- This is the point where the behavior changes.
- Below it, only partner hits matter. Above it, self-hits can matter too.
- Any horizon comparison here is only a comparison, not a finished derivation.

Above wake speed, self-hit feedback can begin



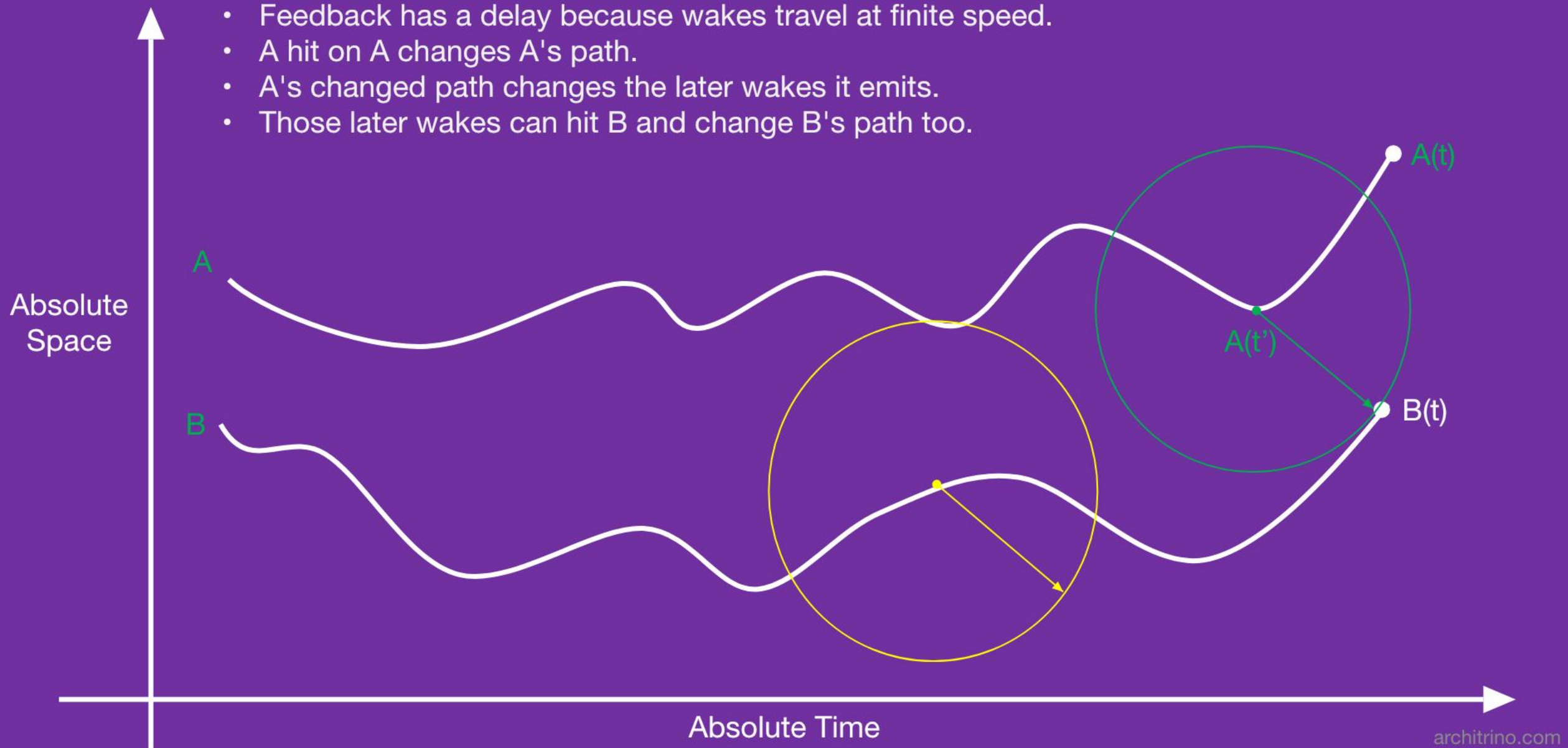
- Above wake speed, an architrino can outrun wakes it emitted earlier.
- If its path crosses one of those older wakes, it receives a self-hit.
- That self-hit can change its later path.

A hit comes from an earlier emission

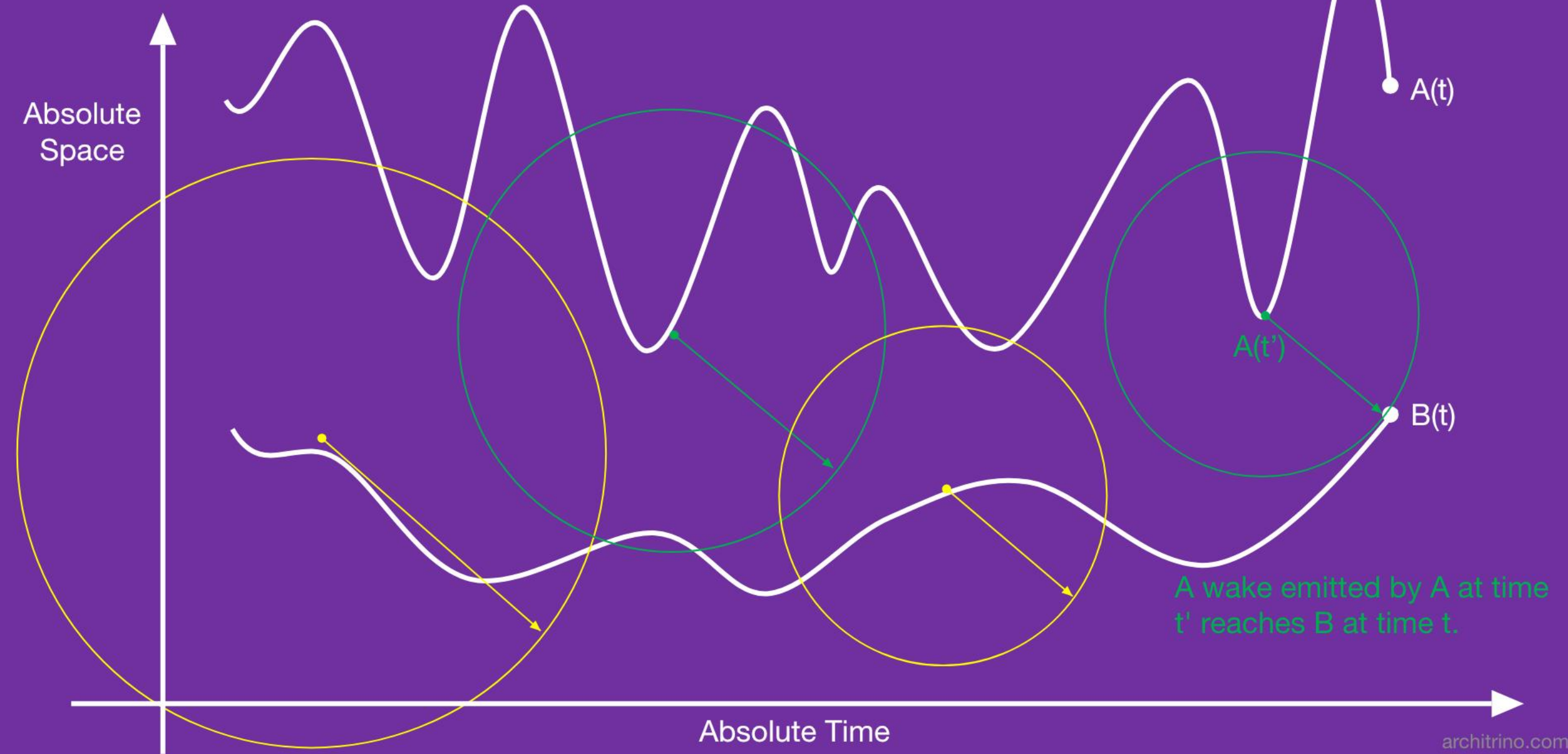


Path changes feed back through later wakes

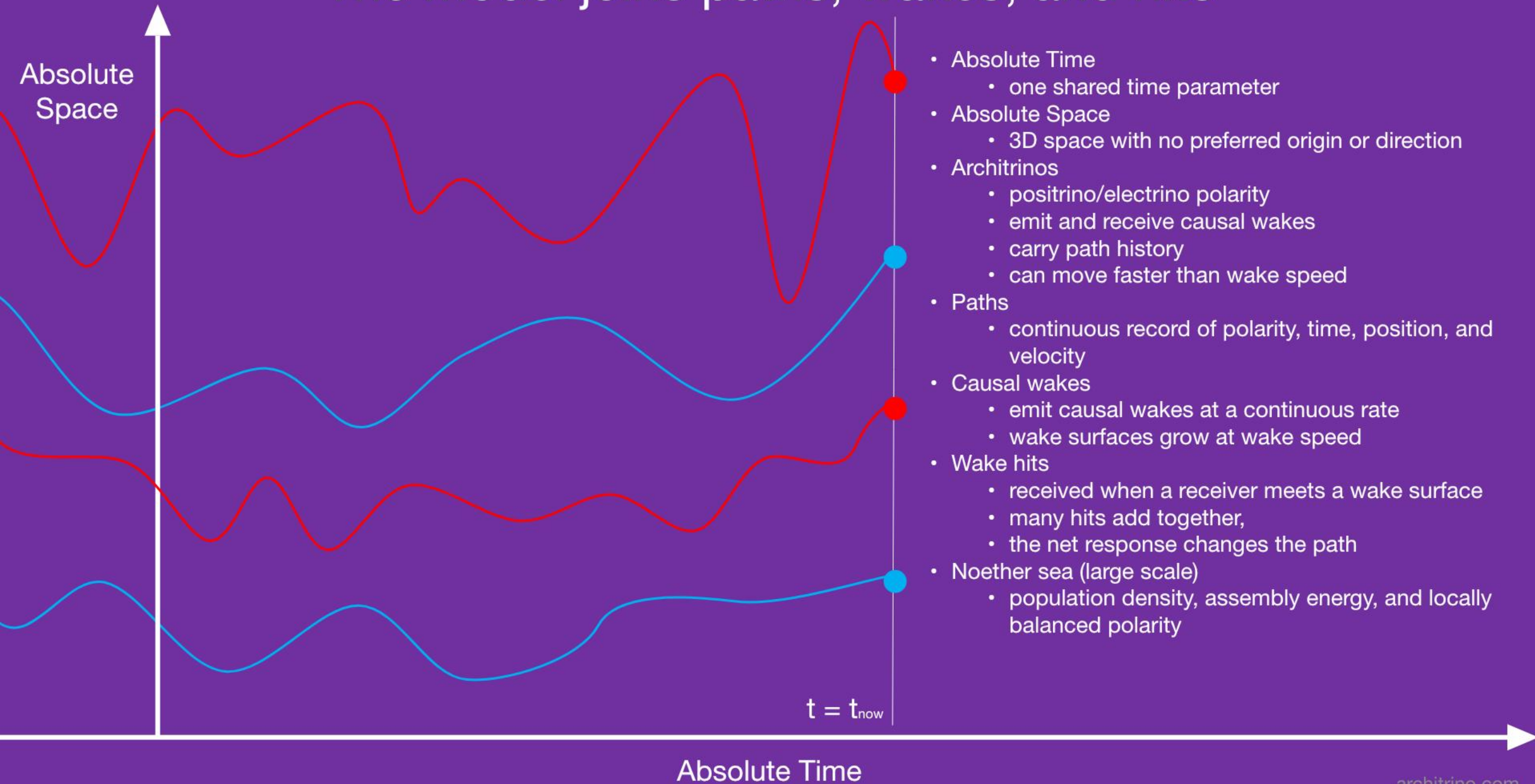
- Feedback has a delay because wakes travel at finite speed.
- A hit on A changes A's path.
- A's changed path changes the later wakes it emits.
- Those later wakes can hit B and change B's path too.



Two paths become coupled through wake history



The model joins paths, wakes, and hits



Zeno's paradox points to the causal-root question

- Zeno's paradox says a trip can be split into endlessly smaller remaining distances.
- That makes motion seem impossible, even though motion clearly happens.
- AAA has a similar trap near a $1/r$ curve:
 - as r approaches 0, the curve seems to demand an infinite response.
 - But the static curve is not the primitive geometry.
- An architrino follows a continuous path, and its causal wakes expand from earlier points on that path at finite wake speed.
- A hit happens only when a receiver crosses a wake surface.
- Near $r = 0$, ask which earlier emissions can reach the receiver (the causal roots), which hits are received, and whether feedback settles, diverges, or becomes invalid.